

Three IDS Models Compared

Three models were trained on the same cleaned CICIDS2018 dataset, with the same 80/20 split, to label each network flow as Benign or one of 14 attack types. This report compares them side by side: how each works, its overall accuracy, how many classes reach 95% recall, and which to use.

Model 1 Single XGBoost 95.7% 10/15 classes at 95% 15-class classifier	Model 2 Two-Level XGBoost 96.3% 9/15 classes at 95% binary gate + namer	Model 3 Isolation Forest + XGBoost 89.1% 4/15 classes at 95% anomaly gate + namer
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Verdict. The **two-level XGBoost wins on accuracy (96.3%)**; the **single XGBoost is the most robust (10/15 classes at 95%)**. The Isolation Forest hybrid is weakest (89.1%, 4/15), because its unsupervised gate drops attacks the others catch. All three hit the same wall on Infiltration and the few-row classes.

01 — HOW EACH MODEL WORKS

Same data, three designs

Model 1 - Single XGBoost. One gradient-boosted model with 15 classes. Each flow gets one prediction directly. Class weights are tuned on a validation set.

Model 2 - Two-Level XGBoost. Level 1 is a binary XGBoost (attack vs not). Flows it calls attack go to level 2, a 14-class XGBoost that names the attack. Both supervised.

Model 3 - Isolation Forest + XGBoost. Stage 1 is an unsupervised Isolation Forest (normal vs attack) that learns only normal traffic; flagged flows go to stage 2, a 15-class XGBoost. Its aim is also to flag unknown attacks.

	Model 1: Single	Model 2: Two-Level	Model 3: IF + XGBoost
Stage 1	-	binary XGBoost	Isolation Forest (unsupervised)
Stage 2	-	14-class XGBoost	15-class XGBoost
Levels	1	2	2
Detects unknown attacks	no	no	yes (stage 1)
Tuning	validation-set class weights	fixed, no validation	threshold (Youden's J)

02 — HEADLINE NUMBERS

Test set of 465,837 flows, unseen in training

Measure	Model 1: Single	Model 2: Two-Level	Model 3: IF + XGBoost
Overall accuracy	95.7%	96.3%	89.1%
Classes at >= 95% recall	10/15	9/15	4/15
Benign recall (normal kept)	95.7%	99.0%	98.6%

Model 3's stage-1 Isolation Forest, judged alone as normal-vs-attack, scores 83.5% normal accuracy, 82.9% attack recall, ROC-AUC 0.865 - fair but not strong, which is why it drops attacks before stage 2 sees them.

03 — PER-CLASS RECALL

How many of 100 attacks each model catches (green = 95%+)

class	test rows	Model 1	Model 2	Model 3
Benign	199,922	95.7%	99.0%	98.6%
DDoS attacks-LOIC-HTTP	115,073	99.9%	100.0%	93.3%
DDoS attack-HOIC	39,772	100.0%	100.0%	82.5%
DoS attacks-Hulk	29,040	100.0%	100.0%	55.8%
Bot	28,907	100.0%	100.0%	99.1%
Infiltration	23,520	51.8%	35.1%	28.0%
SSH-Bruteforce	18,808	100.0%	100.0%	100.0%
DoS attacks-GoldenEye	8,281	100.0%	100.0%	64.2%
DoS attacks-Slowloris	1,982	100.0%	99.9%	99.9%
DDoS attack-LOIC-UDP	346	99.1%	93.9%	93.6%
Brute Force -Web	109	95.4%	70.6%	62.4%
Brute Force -XSS	45	93.3%	97.8%	48.9%
SQL Injection	15	73.3%	46.7%	53.3%
DoS attacks-SlowHTTPTest	9	0.0%	0.0%	11.1%
FTP-BruteForce	8	25.0%	12.5%	0.0%
Overall accuracy	465,837	95.7%	96.3%	89.1%

Model 1 = single XGBoost, Model 2 = two-level, Model 3 = Isolation Forest + XGBoost. Rows sorted by size.

04 — WHAT THE NUMBERS SHOW

- **The big classes are solved by all three.** LOIC-HTTP, HOIC, Hulk, Bot, SSH, GoldenEye and Slowloris are at ~100% for Models 1 and 2. Model 3 drops several of them (Hulk 55.8%, GoldenEye 64.2%, HOIC 82.5%) because its Isolation Forest gate calls those dense floods normal and never passes them on.
- **Model 2 has the best overall accuracy** (96.3%) - its binary gate is a strong, supervised attack/not split, so almost every attack reaches the namer.
- **Model 1 covers the most classes** (10/15) - its validation-set class-weight tuning lifted Brute Force -Web and LOIC-UDP over 95%, which the untuned Model 2 leaves just below.
- **Model 3 is weakest** (89.1%, 4/15). A two-stage system is only as good as its gate, and an unsupervised Isolation Forest cannot separate these attacks from normal traffic (AUC 0.865). Its one advantage - flagging unknown attacks - is not measured here.
- **All three hit the same wall.** Infiltration (looks like normal traffic) and SQL Injection, FTP-BruteForce, DoS-SlowHTTPTest (8-15 test rows after duplicate removal) are below 95% in every model. No model choice fixes these - only merging the tiny classes into families, or accepting them, would.

05 — RECOMMENDATION

If you want...	Use	Why
Highest overall accuracy	Model 2 (two-level)	96.3% overall; strong supervised gate
Most attacks above 95%	Model 1 (single)	10/15 classes; tuned class weights

If you want...	Use	Why
Simplest to run and deploy	Model 1 (single)	one model, one prediction
To flag unknown / novel attacks	Model 3 (IF + XGBoost)	stage 1 flags anything unusual
The tiny classes to pass 95%	none as-is	merge them into families first

Overall: **Model 2 (two-level XGBoost)** for the best accuracy, or **Model 1 (single XGBoost)** for the simplest model with the widest class coverage. Model 3 only if detecting unknown attacks outweighs accuracy on known ones.

Sources: 1_xgboost_model/final_metrics.json, 2_xgboost_model/final_metrics.json, isolation_forest_and_xgboost_model/hybrid_metrics.json & test_metrics.json. Per-model detail is in each folder's own PDF. Compiled 2026-09-29.